P-C-435B
April 5, 1978
SUPERSEDING
Fed. Spec. P-C-435A
July 5, 1972

FEDERAL SPECIFICATION

CLEANING COMPOUND, GENERAL PURPOSE (POWDERED, HEAVY DUTY)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers a heavy duty powdered general purpose cleaning compound suitable for the removal of heavy deposits of oil, grease, and similar soils from hard surface floors such as terrazzo, ceramic tile, marble, and concrete (see 6.1). The product covered in this specification shall be of one type only.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Standard:

Fed. Test Method Std. No. 536/GEN - Soap and Soap Products (Including Synthetic Detergents); Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Philadelphia, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Houston, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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Laws and Regulations:

16 CFR 1500 - Hazardous Substances and Articles, Administration and Enforcement Regulations.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

American Society for Testing and Materials (ASTM) Standards:

D 460 - Sampling and Chemical Analysis of Soaps and Soap Products.
D 820 - Chemical Analysis of Soaps Containing Synthetic Detergents.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

3. REQUIREMENTS

3.1 Material. The compound shall be a uniform, free flowing powder. The compound shall not be harmful to terrazzo, marble, ceramic tile, concrete, or cement surfaces when used according to the recommended dilutions and directions for use.

3.2 Basic requirements. The compound shall conform to the requirements, specified in table I, when tested as specified in table II.

TABLE I

Characteristics	Minimum	Maximum
Matter volatile at 105°C, percent	----	12.0
Anhydrous synthetic detergent (salt-free), percent	4.0	----
Total alkalinity to methyl orange as Na2O, percent	15.0	----
Matter insoluble in water, percent	----	0.3
Matter retained on a no. 10 (2.00 mm) sieve, percent	----	2.0
pH at 25°C, (one percent solution)	9.0	12.0

3.3 Rinsing properties. The compound shall be free rinsing and shall leave no deposit or sediment when tested as specified in 4.4.4.

3.4 Cleaning efficiency. The compound shall demonstrate a cleaning efficiency at least as great as that of the comparison compound specified in the test method described in 4.4.5.

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3.5 Deleterious action. A 0.2 percent (by weight) solution of the compound shall not cause greater than one half the loss of 60-degree specular gloss of the tile surfaces caused by a 0.2 percent solution of trisodium phosphate when tested as specified in 4.4.6.

3.6 Labeling. The container label shall be marked in accordance with the Federal Hazardous Substances Act and normal commercial labeling practice for this type of cleaning compound.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government.

4.2 Inspection sampling. Sampling for the inspection procedures set forth in this specification shall be performed in accordance with MIL-STD-105. The lots to be inspected shall consist of no more than one package size.

4.3 Examination.

4.3.1 Examination of preparation for delivery. An examination shall be made to determine compliance with the requirements of section 5. The sample unit shall be one shipping container fully prepared for delivery. Sampling shall be in accordance with MIL-STD-105. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of percent defective.

4.3.2 Examination of the contents of the filled unit container.

4.3.2.1 Sampling plan. The sample unit shall be defined as one filled unit container. The sample units shall be selected in accordance with inspection level S-2 and the AQL shall be 4.0 percent defective.

4.3.2.2 Examination procedure. The contents of the filled unit container shall be examined for defects in uniformity of product, free flow, labeling, and fill in accordance with the requirements of 3.1, 3.6, and 6.2(c), and shall be scored accordingly.

4.4 Testing.

4.4.1 Sampling plan. One 2-pound sample of the product shall be selected from the lot at random. The lot shall be unacceptable if the sample fails to pass any of the tests described in 4.4.

4.4.2 Testing procedure. The composite sample shall be tested in accordance with the instructions in table II.

TABLE II. Instructions for testing

Characteristics	Specification reference			Results reported as	
	Requirement	Test method		pass or fail	numerically to nearest
Matter volatile at 105°C	Table I	D 460	1/	-	0.1 percent
Anhydrous synthetic detergent	Table I	D 820	1/	-	0.1 percent
Total alkalinity	Table I	D 820	1/	-	0.1 percent
Matter insoluble in water	Table I	D 820	1/	-	0.1 percent
Matter retained on a no. 10 sieve	Table I	2101	2/	-	0.1 percent

1/ Indicates applicable ASTM method.
2/ Indicates applicable method in Federal Test Method Standard No. 536.

4.4.3 pH. Determine to the nearest 0.1 unit the pH of a 1.0 percent solution of the product at 25° ± 2°C using a pH meter.

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4.4.4 Rinsing properties.

4.4.4.1 Synthetic hard water. The synthetic hard water shall be made to the following formula:

0.264 gram/liter $CaCl_2.2H_2O$

0.295 gram/liter $MgSO_4.7H_2O$

The hard water salts should be dissolved in a small portion of distilled water, the solution transferred quantitatively to a liter volumetric flask, and the flask filled to volume.

4.4.4.2 Procedure. Place 2.0 g of the compound and 98 ml of synthetic hard water at room temperature (25° ± 2°C) in a very clean 250 ml Erlenmeyer flask. Stopper the flask and shake vigorously for 1 minute. Pour out the solution. Rinse the flask by the same procedures, using three 75 ml portions of water alone. Invert the flask and allow to dry, and examine for any residue not rinsed from the interior.

4.4.5 Cleaning efficiency. Prepare a comparison standard composed of 23.5 percent Dodecylbenzene sodium sulfonate (85 percent active), 40 percent tripolyphosphate, 7 percent sodium metasilicate (pentahydrate), and 29.5 percent Sodium sulfate (anhydrous). Determine the cleaning efficiencies of 0.5 percent (by weight) solutions of the comparison standard and the cleaning compound sample using the following method.

4.4.5.1 Testing procedure. This procedure shall be carried out five times for each cleaning solution tested, using five separate test panels.

4.4.5.1.1 Panel precleaning. Cut an 18 inch piece from the supply of cove base (see 4.4.5.3.5) to fit the washing tray of the Gardner Washability machine (see 4.4.5.3.2). The 18 inch panel shall then be scrubbed by hand using a cellulose sponge (see 4.4.5.3.7) saturated with a solution consisting of a 0.5 wt percent solution of linear alkylaryl sodium sulfonate (85 percent active) (see 4.4.5.3.1) in tap water. Rinse the panel under cool running tap water and then drain for 10 minutes. Blot the remaining water with a paper towel. Oven dry for 15 minutes at 100°C and then allow to cool at room-temperature. Determine the reflectance of the panel (see 4.4.5.2) and record.

4.4.5.1.2 Panel soiling.

4.4.5.1.2.1 Preparation of soiling mixture.

Soiling mixture (see 4.4.5.3)	Parts by weight
Metallic brown	20
Odorless kerosene	12
Stoddard solvent	12
White mineral oil (UPS)	1
Lubricating oil (SAE 10)	1
Hydrogenated vegetable shortening	1

Combine oils and shortening and heat to 41° ± 1°C to melt. Add half the kerosene. With a spatula, mix in the metallic brown. Add remaining kerosene and Stoddard solvent, and mix for 2 hours with a magnetic stirrer.

4.4.5.1.2.2 Soiling procedure.

While the bulk lot of soil is continuously stirred to insure uniformity, a sample of approximately 5 g is removed and applied along entire length of the vinyl panel. A Byrd applicator or "Doctor" blade (see 4.4.5.3.4), set at 0.008 inch clearance and 2 inches width, is used to provide an even spread.

After soil application, air-dry panels at least 30 but not more than 60 minutes, then cure panels on smooth surface in a forced draft oven at 100° ± 3°C for 60 minutes. Remove the panels from oven and cool to room-temperature, using the panels within 24 hours.

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4.4.5.1.3 Panel washing.

4.4.5.1.3.1 Preparation of test detergent. Synthetic hard water shall be prepared according to the following formula:

0.132 gram $CaCl_2.2H_2O$ (CP).

0.1475 gram $MgSO_4.7H_2O$ (CP).

Accurately weigh the salts and dissolve them in a small portion of distilled or deionized water, quantitatively transfer them to a 1 liter volumetric flask and make up to volume. Dry detergents shall be used in this hard water at 0.5 weight percent concentrations or at a recommended concentration, and liquid products shall be tested at the recommended use concentration.

4.4.5.1.3.2 Washing procedure. The sponge (see 4.4.5.3.7) is cut to size when dry to fit the sponge box. The thickness of the sponge shall be 1 inch when wet. Total weight of the box and dry sponge shall be 1 pound ± 1/2 ounce. Soak panel for 60 seconds in wash solution sufficient to cover entire panel. Wet sponge with 25° ± 3°C tap water and squeeze damp-dry. Add 50 ml of detergent solution to sponge and insert sponge in box. Center test panel in washability pan with two properly sized pieces of vinyl molding. Start apparatus set at one stroke per second (one cycle back and forth is two strokes), while dripping detergent solution from a pipet onto test panel center at a rate of 12 ml of solution during the wash period of 100 strokes. Stop apparatus, remove and rinse test panel under a light stream 25° ± 3°C tap water. Drain and replace panel in apparatus in the reverse direction. Rinse sponge in 25° ± 3°C tap water, squeeze damp-dry, add 50 ml of detergent solution to unused side, replace sponge in box and repeat 100 stroke wash cycle as before. Rinse panel and air dry. Determine the reflectance (see 4.4.5.2.1) of the washed portion of the panel and record.

4.4.5.2 Evaluation procedure.

4.4.5.2.1 Reflectance measurement. Using a photometer (see 4.4.5.3.3) and search unit equipped with tristimulus green filter, reflectance of the surface of the test panel is measured before soiling and after washing using a vinyl panel template of the dimensions given in figure I. Four reflectance measurements shall be made of different areas of the soiled and washed portion of each panel examined. The four readings are measured to the nearest 0.5 percent, and averaged. This procedure shall be repeated for each of the five panels required to test each cleaner. Between panels, the search unit is rested on a standard white plaque of approximately 80 percent reflectance and the instrument checked before each series of readings.

4.4.5.2.2 Cleaning efficiency (C.E.)

$$\text{Percent cleaning efficiency} = \frac{R_2}{R_1} \times 100$$

Where R_1 = average reflectance of unsoiled, precleaned panel.
R_2 = average reflectance of soiled, test washed panel.

Report the cleaning efficiency results as the average of the cleaning efficiencies obtained from the five test panels. The results shall be reported to the nearest unit percent.

4.4.5.3 Special apparatus and materials.

4.4.5.3.1 Reagents.

Sodium Metasilicate, Pentahydrate (Technical $Na_2SiO_3.5H_2O$)

Suppliers:

Philadelphia Quartz Co.
1123 Public Ledger Bldg.
Philadelphia, PA 19106

Diamond Shamrock Chemical Co.
300 Union Commerce Bldg.
Cleveland, OH 44115

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Sodium Tripolyphostphate, Granular ($NA_5P_3O_10$)

Suppliers: Monsanto Company
800 N. Lindbergh Blvd.
St. Louis, MO 63166

FMC Corporation
Inorganic Chemicals Division
633 3rd Avenue
New York, NY 10017

Linear dodecylbenzene sodium sulfonate (85 percent active)

Supplier: Witco Chemical Corporation

Metallic Brown

Supplier: General Services Administration
Federal Supply Service
Chemicals Branch (FMBC)
Washington, DC 20406

Hydrogenated Vegetable Shortening

Kerosene, Odorless

Lubricating Oil (SAE 10)

Sodium Sulfate, Anhydrous

Stoddard Solvent

White Mineral Oil (USP)

4.4.5.3.2 Straightline Washability Apparatus (Henry A. Gardner Laboratory, Inc., Bethesda, MD 20014).

4.4.5.3.3 Photovolt Model 670 or equal with a 610Y search unit and tristimulus green filter (range 520 to 540 millimicrons) manufactured by Photovolt Corp., 1115 Broadway, New York, NY 10010 has been found suitable.

4.4.5.3.4 Gardner adjustable blade ("doctor" blade) Model AG 3820 A (Henry A. Garnder Laboratory, Inc., Bethesda, MD 20014).

A substitute for the "doctor" blade is the Byrd Applicator AR-3800 of 2 inch width and a clearance of 8 mils (Henry A. Gardner Laboratory, Inc., Bethesda, MD 20014).

4.4.5.3.5 White Vinyl Cove Base.

General Services Administration
Federal Supply Service
Chemicals Branch (FMBC)
Washington, DC 20406

4.4.5.3.6 Forced draft oven having $\pm$ 2°C tolerance.

4.4.5.3.7 Cellulose sponge, photographic grade (fine pore).

4.4.5.3.8 Standard white plaque (reflectance approximately 80 percent).

4.4.5.3.9 Template of vinyl panel, 2 inches by 4 inches with four evenly spaced holes for making reflectance readings (see figure I).

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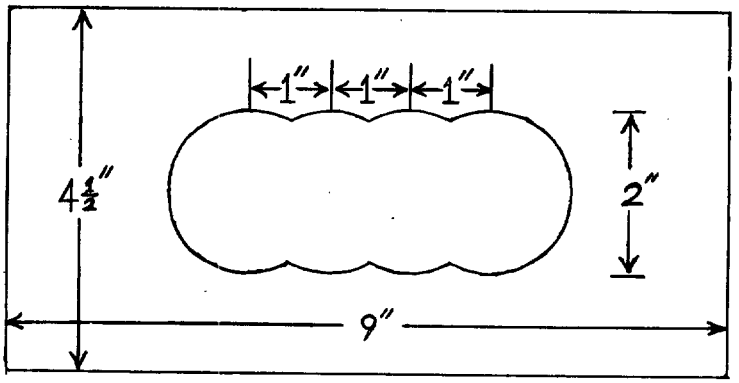


FIGURE I. Template dimensions

4.4.6 Deleterious action.

4.4.6.1 Test tile. The test tiles used in this test shall be white terrazzo tile (Demarco white) made with 50 percent no. 1 chip, 50 percent no. 2 chip, and white portland cement. The tile can be obtained in 4 in x 4 in x 5/8 in from the American Terazzo Co., 5607 Dyer Street, Dallas, TX 75206. The tiles shall be obtained polished, but without sealer.

4.4.6.2 Procedure. Prepare 1 liter each of a 0.2 percent (by weight) solution of trisodium phosphate (sodium phosphate, tribasic) and a 0.2 percent solution (by weight) of the cleaning compound under test in distilled water. Measure the 60 degree specular gloss (using a Gardner Glossmeter or equivalent) of two panels and then immerse one panel in each of these solutions, kept at room temperature, for 1 hour. Remove the panels, rinse thoroughly in distilled water, and dry at room temperature for at least 1 hour. Again measure the specular gloss of each panel. Determine the percentage decrease in gloss of the two panels.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A or commercial, as specified (see 6.2).

5.1.1 Level A. The containers shall conform to the requirements of the Uniform or National Motor Freight Classification.

5.1.1.1 Twenty-five pound quantity. Twenty-five pounds of cleaning compound shall be packed in a pail. The cover and pail shall neither affect nor be affected by the product. When metal components are furnished they shall be lined or coated with a material that will neither affect nor be affected by the product. The pail shall be securely closed.

5.1.1.2 One-hundred and four-hundred pound quantities. One-hundred or four-hundred pound quantities shall be packed in a fiber drum. The drum shall be fitted with a polyethylene bag liner not less than 0.002 inch (2 mils) thick. The bag and drum shall be securely closed to prevent sifting of the product.

5.1.2 Commercial. The cleaning compound shall be packed in shipping containers to insure safe delivery at destination, to provide for safe redistribution by the initial receiving activity, and shall be acceptable by common carrier under the National Motor Freight Classification or Uniform Freight Classification.

5.2 Palletization. When shipments to Government depots are full car or truck-load, the shipping containers shall be palletized on expendable pallets, 2 or 4 way design, to facilitate handling in accordance with normal commercial practice. The palletized load shall not exceed 2,500 pounds in weight, 63 inches in height, 56 inches in length, and 45 inches in width.

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5.3 Marking. Marking shall be as specified in the contract or order.

6. NOTES

6.1 Intended use. The cleaning compound covered by this specification is intended for use in the cleaning of terrazzo, travertine, marble, and ceramic flooring and for the removal of heavy deposits of oil, grease, and soil in shops, garages, hangars, and ramps. Not for use on resilient floors.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Packing and quantity per pack required (see 5.1).
(c) Marking required (see 5.3).

Military Custodians:

Army - GL (MCA)
Air Force - 99

Reviewer activity:

Army - MD

User activity:

Navy - MC

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA - PBO
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